BioOne DIGITAL LIBRARY

Entomopathogenic Nematodes and Fungi Virulence to Cowpea Curculio (Coleoptera: Curculionidae) Larvae1

Authors: Schmidt, Jason M., Shapiro-Ilan, David I., Graham, Casey, Barwick, Sydni, Sparks, Alton, et al.

Source: Journal of Entomological Science, 53(2) : 152-161

Published By: Georgia Entomological Society

URL: https://doi.org/10.18474/JES17-65.1

The BioOne Digital Library (https://bioone.org/) provides worldwide distribution for more than 580 journals and eBooks from BioOne's community of over 150 nonprofit societies, research institutions, and university presses in the biological, ecological, and environmental sciences. The BioOne Digital Library encompasses the flagship aggregation BioOne Complete (https://bioone.org/subscribe), the BioOne Complete Archive (https://bioone.org/archive), and the BioOne eBooks program offerings ESA eBook Collection (https://bioone.org/esa-ebooks) and CSIRO Publishing BioSelect Collection (https://bioone.org/csiro-ebooks).

Your use of this PDF, the BioOne Digital Library, and all posted and associated content indicates your acceptance of BioOne's Terms of Use, available at www.bioone.org/terms-of-use.

Usage of BioOne Digital Library content is strictly limited to personal, educational, and non-commmercial use. Commercial inquiries or rights and permissions requests should be directed to the individual publisher as copyright holder.

BioOne is an innovative nonprofit that sees sustainable scholarly publishing as an inherently collaborative enterprise connecting authors, nonprofit publishers, academic institutions, research libraries, and research funders in the common goal of maximizing access to critical research.

Downloaded From: https://bioone.org/journals/Journal-of-Entomological-Science on 19 Mar 2026
Terms of Use: https://bioone.org/terms-of-use Access provided by Clemson University

Entomopathogenic Nematodes and Fungi Virulence to Cowpea Curculio (Coleoptera: Curculionidae) Larvae[1]

Jason M. Schmidt[2], David I. Shapiro-Ilan[3], Casey Graham, Sydni Barwick, Alton Sparks Jr., and David Riley

Department of Entomology, College of Agricultural and Environmental Sciences, University of Georgia, Tifton Campus, 2360 Rainwater Road, Tifton, Georgia 31793-5766 USA

J. Entomol. Sci. 53(2): 152–161 (April 2018)

Abstract Cowpea curculio, *Chalcodermus aeneus* (Boheman), is the major pest of southern peas or cowpea, *Vigna unguiculata* (L.). Throughout the southeastern United States, current management recommendations for cowpea curculio rely on insecticide applications; however, resistance has been observed. Entomopathogenic biocontrol agents, specifically nematodes in the genera *Steinernema* and *Heterorhabditis* and fungi in the genera *Beauveria* and *Metarhizium*, have shown efficacy with other curculio pests. We conducted laboratory bioassays to assess the virulence of entomopathogenic agents as a first step to provide alternative control strategies against cowpea curculio. Using controlled environmental conditions, we tested the effects of nematodes (*Steinernema carpocapsae* Weiser, *Steinernema feltiae* Filipjev, *Heterorhabditis bacteriophora* Poinar, and *Heterorhabditis indica* Poinar) and two fungi (*Beauveria bassiana* Balsamo and *Metarhizium brunneum* Petch) on survival and infectivity of cowpea curculio. *Heterorhabditis indica* had the greatest effect on cowpea curculio larvae, resulting in less than 20% survival and greater than a 65% infection rate. Conversely, cowpea curculio larvae exposed to fungal pathogens had similar survival to larvae exposed to water and less than 20% infection rates. For fungal pathogens, applying the pathogen to the soil surface prior to introduction of cowpea curculio larvae led to infection of the larvae. Our results suggest nematodes and fungal pathogens are capable of infecting cowpea curculio. Our data provide the basis for further tests examining the timing and application rates in field situations.

Key Words biological control, *Beauveria*, cowpea curculio, *Heterorhabditis*, *Metarhizium*, *Steinernema*, *Vigna unguiculata*

The cowpea curculio, *Chalcodermus aeneus* (Boheman), is the major pest of southern peas or cowpea, *Vigna unguiculata* (L.), throughout the southeastern United States. The insect has a life cycle from egg to adult that requires approximately 30–40 d (Arant 1938). Adults emerge from overwintering sights in mid-May from noncrop plants and past fields of cowpea to infest cowpea (Sudbrink et al. 1998). A second generation occurs with peak adult activity in August–September. Adults oviposit into the bean pods where larvae consume

[1]Received 17 May 2017; accepted for publication 24 July 2017.
[2]Corresponding author (email: jschmid2@uga.edu).
[3]U.S. Department of Agriculture, Agricultural Research Service, Southeastern Fruit and Tree Nut Research Laboratory, 21 Dunbar Road, Byron, GA 31008, USA.

Downloaded From: https://bioone.org/journals/Journal-of-Entomological-Science on 19 Mar 2026
Terms of Use: https://bioone.org/terms-of-use Access provided by Clemson University

beans as energy for growth (Arant 1938; Riley et al. 2014; Sudbrink et al. 1998). Larval development occurs within the beans, and fourth instars drop to the soil where they burrow to a depth of 2.54–7.62 cm, form a puparium, and complete their development. Adults emerging in late fall enter protective sights for overwintering (Arant 1938). Recent research suggests a small percentage of the fall population may overwinter in the soil (A.S. and D.R., unpubl. data), likely as pupae or adults.

Management recommendations for cowpea curculio rely on scheduled, preventive insecticide application from first bloom (or pin-sized pods) to harvest (Riley et al. 2014, 2017). As resistance issues emerged in the 1990s, organophosphate insecticides initially used for management were replaced by pyrethroid insecticides (Nguessan and Chalfant 1990). Cowpea curculio, however, had already exhibited reduced response to a pyrethroid, esfenvalerate, in the late 1980s (Nguessan and Chalfant 1990). Recent efficacy trials and field experience indicate pyrethroid insecticide resistance is severe and widespread in this species, leaving growers with no efficacious insecticides registered for management of this pest. Therefore, research on developing alternative control strategies is warranted.

Of considerable research interest for agricultural pest management are entomopathogenic nematodes in the genera *Steinernema* and *Heterorhabditis* (Poinar and Grewal 2012; Shapiro-Ilan et al. 2012). In nature, both species are insect obligate parasites (Lewis and Clarke 2012; Poinar et al. 1990). Their mode of action is due to a mutualistic relationship with bacterial species found in the bodies of the nematodes. Each nematode species is associated with one bacteria species; *Xenorhabdus* spp. are associated with steinernematids and *Photorhabdus* spp. with heterorhabditids (Poinar et al. 1990). Effective control of several weevil species (Coleoptera: Curculionidae) has been shown with entomopathogenic nematodes (Nagesh et al. 2016; Shapiro-Ilan and Cottrell 2006; Shapiro-Ilan et al. 2004; Williams et al. 2015,); however, study on their efficacy on cowpea curculio has not been conducted.

Entomopathogenic fungi are soil-borne microbes that exhibit efficacy for controlling a variety of insect pests. Two promising genera of fungi with broad efficacy against insect hosts are *Beauveria* spp. and *Metarhizium* spp. (Jaramillo et al. 2005; Shapiro-Ilan et al. 2011a; Wraight and Ramos 2002). Entomopathogenic fungi exhibit potential as a soil treatment for larval control against the cowpea curculio in South Carolina (USA) and a different species of *Chalcodermus* in Brazil (Bell and Hamalle 1970; Quintella and Roberts 1992).

To evaluate nonpesticide control strategies for cowpea curculio, we tested susceptibility of cowpea curculio larvae to entomopathogens including commercially available representatives of two genera of nematode (*Steinernema* and *Heterorhabditis*, Poinar) and fungi (*Beauveria*, Balsamo and *Metarhizium*, Petch). Nematodes exhibit sensitivity to desiccation and ultraviolet radiation (Kaya and Stock 1997; Lacey and Shapiro-Ilan 2008) and so we targeted below-ground stages for initial evaluation. We hypothesized entomopathogenic nematodes and fungi exhibit different virulence patterns on cowpea curculio, and virulence may be influenced by application rate.

Downloaded From: https://bioone.org/journals/Journal-of-Entomological-Science on 19 Mar 2026
Terms of Use: https://bioone.org/terms-of-use Access provided by Clemson University

Table 1. Experimental rates of bioassay treatments used to evaluate the virulence of different entomopathogens for cowpea curculio, *C. aeneus.*

Entomopathogen	Treatment	Application Rate
Control (con); no entomopathogens	Water	na
Heterorhabditis bacteriophora (VS)	Low/high	250 IJs/1000 IJs
Heterorhabditis indica (HOM1)	Low/high	250 IJs/1000 IJs
Steinernema carpocapsae (All)	Low/high	250 IJs/1000 IJs
Steinernema feltiae (SN)	Low/high	250 IJs/1000 IJs
Beauveria bassiana (GHA)	Low/high	250K spores/4M spores
Metarhizium brunneum (F52)	Low/high	250K spores/4M spores

na = not applicable

Materials and Methods

We collected fourth-instar cowpea curculio larvae from infested peas for immediate use by placing larvae on large mesh screens that enabled them to drop into a plastic container. All nematodes and fungi used in our bioassays were reared at the U.S. Department of Agriculture–Agricultural Research Service (USDA–ARS), Southeastern Fruit and Tree Nut Laboratory in Byron, GA. *Heterorhabditis bacteriophora* Poinar (VS strain), *Heterorhabditis indica* Poinar, Karunakar and David (HOM1 strain), *Steinernema carpocapsae* (Weiser) (ALL strain), and *Steinernema feltiae* (Filipjev) (SN strain) were cultured on *Galleria mellonella* (L.) larvae at 25°C according to Kaya and Stock (1997). We cultured *Beauveria bassiana* (Balsamo) Vuillemin (GHA strain) and *Metarhizium brunneum* Petch (F52 strain) on potato dextrose agar (Goettel and Inglis 1997).

Soil used for the bioassays was collected from the University of Georgia (UGA) Horticulture Farm, Tifton, GA. Soil samples were analyzed at the UGA Soil, Plant and Water Laboratory in Athens. Soil was a loamy sand containing, on average, 181 parts per million (ppm) calcium carbonate, 411 ppm calcium, 40 ppm potassium, 38 ppm magnesium, 6 ppm manganese, 55 ppm phosphorus, and 2 ppm zinc, with a pH of 5 and composed of, on average, 83% sand, 11% silt, and 7% clay. All experiments were conducted at the USDA–ARS Southeastern Fruit and Tree Nut Laboratory, Byron, GA.

Plastic soufflés cups (36.9 mL) served as bioassay arenas based on procedures described by Shapiro-Ilan et al. (2003). We placed oven-dried, autoclaved soil (27.5 g) in each arena and standardized final moisture levels to field capacity using 17% filtered water. In 2015, treatments consisted of three nematode strains: *H. indica* (HOM1), *S. carpocapsae* (ALL), *S. feltiae* (SN), and two fungal strains, *B. bassiana* (GHA) and *M. brunneum* [F52]). An untreated water-only was included as a control (Table 1). In 2016, *H. bacteriophora* (VS) was substituted for HOM1, and a second experiment was conducted to compare VS to HOM1 (Table 1). Microbial treatments

Downloaded From: https://bioone.org/journals/Journal-of-Entomological-Science on 19 Mar 2026
Terms of Use: https://bioone.org/terms-of-use Access provided by Clemson University

were standardized and applied at two application rates, high and low, for each species (Table 1). All cultures were at minimum 95% viable at time of application. Experimental application rates were based on results of a preliminary concentration gradient experiment (D.I.S-I. et al. unpubl. data). All concentrations of entomopathogens were prepared by serially diluting stock suspensions of either infective juveniles (IJs) or fungal conidia.

An individual cowpea curculio larva was placed in each arena and the appropriate agent and concentration was added to the soil in a 0.5-mL droplet. Arenas were maintained at 25°C in an environmental chamber and survival and infection status was recorded after 7 d. Infection status was determined by the proportion of insects exhibiting signs of infection according to Kaya and Stock (1997) for nematode treatments and Goettel and Inglis (1997) for the fungal treatments. Larval mortality was determined by a lack of response, signs of infection, or putrefaction.

To determine the effect of microbial type (nematode vs. fungi) and application rate (low vs. high) on larval survival and infection, we used generalized linear mixed-effects models (log-link function using (glmer{lme4}) for analysis of binary data (Pinheiro and Bates 2000; R Core Team 2016). Each experiment was conducted in a completely randomized design with 27 replicates per treatment. The random portion of the model was specified as the replicate nested within a trial for a given treatment. We treated 2015 and 2016 experiments as separate analyses. For significant main effects, we conducted pair-wise multiple comparisons with adjusted general linear comparisons using glht (Hothorn et al. 2008).

Results

In 2015 ($n = 230$ larvae), nematode and fungal application treatments had a significant effect on survival of cowpea curculio larvae ($\chi^2 = 24.13$, $df = 5$, $P = 0.0002$). Application rate had a marginal influence on survival ($\chi^2 = 2.56$, $df = 1$, $P = 0.1096$), and there was a nonsignificant interaction between nematode and fungal treatment and application rate ($\chi^2 = 5.81$, $df = 4$, $P = 0.2142$; Fig. 1A). Nematode species had the greatest effect on survival. The lowest proportion of larvae remaining alive after 7 d occurred in the high application rate of the nematode *H. indica*, where 0 curculio larvae survived (HOM1 treatment; Fig. 1A). Moreover, infection was only observed in treatments where nematodes were present ($\chi^2 = 259.21$, $df = 10$, $P < 0.0001$; Fig. 1B).

Trials conducted in 2016 ($n = 264$ larvae) showed an overall similar pattern of cowpea curculio larvae infection by nematodes and fungi as the 2015 data (Fig. 1D). Cowpea curculio larval survival was significantly influenced by treatments ($\chi^2 = 26.49$, $df = 5$, $P < 0.0001$), application rate ($\chi^2 = 7.92$, $df = 1$, $P = 0.0049$), and a significant interaction ($\chi^2 = 12.69$, $df = 4$, $P = 0.0128$). The interaction is explained by the proportion alive in the high application rate for *B. bassiana* being higher than expected (Fig. 1C). Nematodes had higher infection rates than did fungal pathogens, and *H. bacteriophora* had a lower infection rate than the other two nematode species tested (Fig. 1D).

Despite overall similarities between 2015 and 2016 data, larvae were collected at different times during the season due to seasonal availability and a different

Downloaded From: https://bioone.org/journals/Journal-of-Entomological-Science on 19 Mar 2026
Terms of Use: https://bioone.org/terms-of-use Access provided by Clemson University

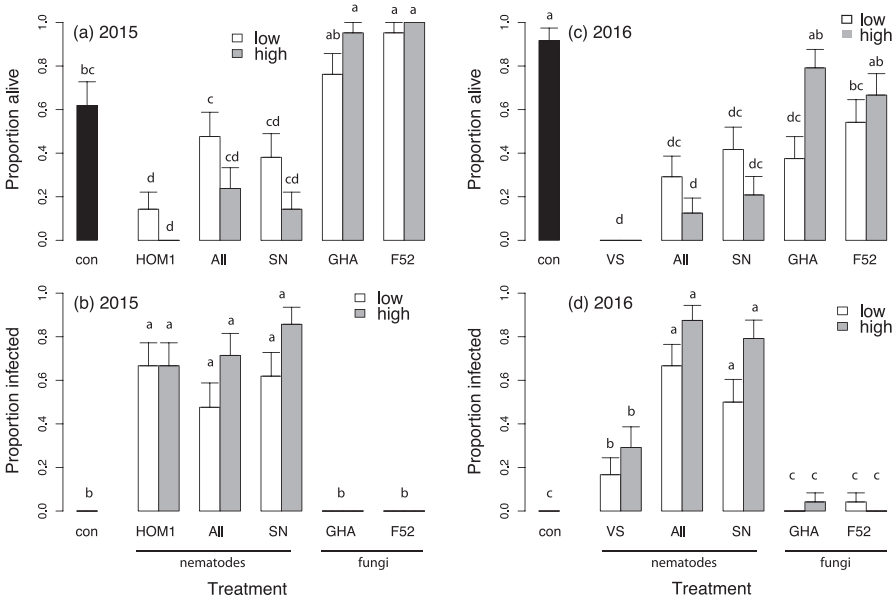

Fig. 1. Survival and infection of cowpea curculio, *C. aeneus*, larvae 7 d following exposure to nematodes and fungi in laboratory bioassays using *H. indica* (HOM1), *H. bacteriophora* (VS), *S. carpocapsae* (ALL), *S. feltiae* (SN), *B. bassiana* (GHA), and *M. brunneum* (F52). Letters indicate the adjusted linear multiple comparisons made between treatments with dissimilar lettering to indicate significantly different treatments ($\alpha = 0.05$).

species of nematode was used. Cowpea curculio larvae were collected in fall for 2015 experiments whereas larvae emerging in summer were used in 2016. Another major difference between the 2 yr was the use of different strains, *H. indica* in 2015 and *H. bacteriophora* in 2016, with *H. indica* unavailable in culture at the time in 2015 larvae emergence and VS being available during cowpea curculio emergence in 2016. *Heterorhabditis indica* and *H. bacteriophora* appear to provide the greatest efficacy in their respective studies, so a subsequent experiment ($n = 420$ larvae) was conducted to compare *H. bacteriophora* to *H. indica*. Significant treatment effects were detected due to nematode treatments showing lower survivorship over the 7-d period compared to the control ($\chi^2 = 42.26$, $df = 4$, $P < 0.00001$; Fig. 2). In addition, the low application rate of *H. indica* had higher survival than the high application rate for *H. bacteriophora* and, numerically, the *H. bacteriophora* had the lowest survival rate (Fig. 2).

Discussion

Here we present the first evaluation of susceptibility of *C. aeneus*, the cowpea curculio, to entomopathogenic nematodes. Nematode virulence toward larval

Downloaded From: https://bioone.org/journals/Journal-of-Entomological-Science on 19 Mar 2026
Terms of Use: https://bioone.org/terms-of-use Access provided by Clemson University

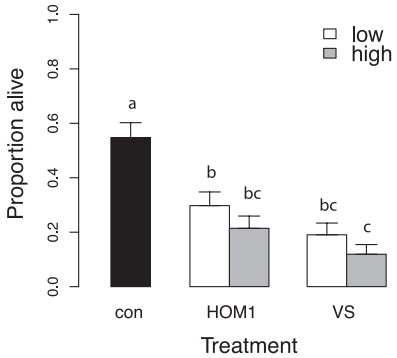

Fig. 2. Survival and infection of cowpea curculio, *C. aeneus*, larvae 7 d following exposure to *H. indica* (HOM1) vs. *H. bacteriophora* (VS). Letters indicate the adjusted linear multiple comparisons made between treatments with dissimilar lettering to indicate significantly different treatments ($\alpha = 0.05$).

stages of cowpea curculio was related to the species tested, with nematodes exhibiting higher virulence levels than did fungi. Fewer surviving larvae were observed following nematode exposure, and higher levels of infection were detected for nematodes as compared to fungal species. Past studies for evaluating nematode virulence on other weevil species showed similar levels of virulence between species (Shapiro-Ilan et al. 2011b; Smith et al. 1993) and, in other host species, differential susceptibility (Poinar and Grewal 2012; Stark and Lacey 1999). For the fungal species tested, curculio larval survival did not differ greatly from larvae exposed to water, and few larvae showed any symptoms of being infected by fungal entomopathogens.

This is the first study to document virulence of common entomopathogenic nematodes for cowpea curculio control. Studies have shown differences in nematode virulence in other curculionid species (e.g., Mannion and Jansson 1992; Shapiro-Ilan and McCoy 2000; Shapiro-Ilan et al. 2011b), as was detected with cowpea larvae in our tests. High virulence levels and infection rates were observed for nematodes used from two genera: *Steinernema* and *Heterorhabditis*. In the present study, we tested two species of *Steinernema* (*S. carpocapsae* and *S. feltiae*) and two species of *Heterorhabditis* (*H. indica* and *H. bacteriophora*); these have been tested in multiple systems to show a high degree of efficacy for biological control of soil-inhabiting target pests. For cowpea curculio, any one of these species appears to have potential to reduce survival considerably with high infection rates, which would lead to further dispersal of nematodes in the soil. This needs to be evaluated under field conditions.

Fungal entomopathogens performed poorly in bioassays against cowpea curculio. Reasons might include concentration of fungal spores used in the bioassay. Bell and Hamalle (1970) used conidial concentrations more than an order of magnitude greater than the rates used in our study. Quintella and Roberts (1992) also found very high conidia rates were needed to achieve infection of *Chalcodermus bimaculatus* (Boheman) (Coleoptera: Curculionidiae), a congener

Downloaded From: https://bioone.org/journals/Journal-of-Entomological-Science on 19 Mar 2026
Terms of Use: https://bioone.org/terms-of-use Access provided by Clemson University

of cowpea curculio. In our study the bioassay methodology may have been a contributing factor because the cowpea curculio larvae were introduced and burrowed quickly into the soil, followed by addition of entomopathogens at soil level. Although some interaction of fungal spores with larvae occurred, separation of host and pathogen may have impacted on the efficacy of immobile fungal spores, whereas nematodes are mobile and could more easily reach the larvae (Dara et al. 2017). We conducted a small, subsequent experiment using the same conidia rates as presented in Table 1 to assess if a fungal barrier applied prior to larval introduction would alter our results. Percent survival observed in treatments containing *B. bassiana* was 6.5% and in *M. brunneum* was 12.5%, which are significantly lower than the control (43.75%; $n = 14$/treatment; control vs. *B. bassiana*, estimated z-value $= -2.35$, $P = 0.042$; and control vs. *M. brunneum,* estimated z-value $= -2.35$, $P = 0.042$). Cowpea larvae in fungal treatments displayed symptoms of infection, and any individuals dying in the control treatment had undetermined causes of death. Therefore, success of using *B. bassiana* or *M. brunneum* may depend on application to foliage or the soil surface prior to larvae dropping to the soil.

Control mortality was relatively high in all of our trials. A high level of control mortality with curculionid species is common in laboratory assays, as there are numerous natural pathogens or other factors that may influence survivorship (Shapiro-Ilan 2001; Shapiro-Ilan et al. 2003, 2011a). We observed higher survival in treatments with *B. bassiana* or *M. brunneum* as compared to control treatments exposed to only water when larvae were introduced before fungal applications. Control mortality may be due to interactions of curculionid with endemic bacteria (Shapiro-Ilan et al. 2004; Slatten and Larson 1967), larval age, year or season (Bell and Hamalle 1971), or a dipterous parasitoid (Slatten and Larson 1967). Antagonistic interactions with endemic microbes (e.g., *Serratia marcescens*) may hinder insect infection by entomopathogens added to the system (Shapiro-Ilan et al. 2004). Such interactions may have impacted results; however, we did not isolate bacteria from the larvae during our experiment and we found no evidence of dipterous parasites. The age of instars was controlled as best as possible, and individuals were collected immediately after falling from beans into trays and used in experiments within 1 d. Further study is needed to determine the time of year for larvae drop. There are two population cycles of *C. aeneus* recorded during the season (D.R. unpubl. data) and, to provide enough material, we used larvae from both times of the year. This may have increased the variation in our data, as individuals may be more susceptible to infection at some periods of the year or potentially carrying a parasitic bacteria (Shapiro-Ilan 2001).

The knowledge gap we targeted in our study was understanding which readily available entomopathogens show efficacy for soil life stages of cowpea curculio. Based on our results, we conclude multiple species of nematodes and fungal species show virulence for the cowpea curculio. We determined that relatively low rates of nematodes influenced cowpea survival, which is important for a crop such as the southern pea because application cost is a deterrent for using this type of biocontrol. Fungal species were only effective when applied to soil cups prior to adding the larvae, which would expose larvae to condia prior to entering the soil. Our results provide preliminary information on the potential importance of application timing of fungal control for larvae. Application timing also has the

Downloaded From: https://bioone.org/journals/Journal-of-Entomological-Science on 19 Mar 2026
Terms of Use: https://bioone.org/terms-of-use Access provided by Clemson University

potential to reduce additional damage of pods by infecting the adult stage. Further research is needed on adults, as we have observed natural fungal growth on cowpea curculio adults (A.S. and D.R. unpubl. data). To our knowledge, the use of such a broad range of entomopathogens for control of cowpea curculio has not been previously studied, and our results provide the basis for future studies and the possible integration of entomopathogens in cowpea curculio suppression.

Acknowledgments

A special thanks to Stacy Byrd for assistance with laboratory bioassays and Melissa Thompson in coordinating laboratory studies and soil sterilization. We thank the University of Georgia for partial funding for this initial work as part of the start-up package for J.M.S. We thank two anonymous reviewers and the editor of Journal of Entomological Science who provided valuable comments contributing to the final presentation. A Southern Region Integrated Pest Management grant (U.S. Department of Agriculture–National Institute of Food and Agriculture 15-0085-19) supported discussions of work and planning. Any opinions, findings, conclusions, or recommendations expressed in this publication are those of the author(s) and do not necessarily reflect the view of the U.S. Department of Agriculture.

References Cited

Arant, F.S. 1938. The relative efficiency of certain fluorine and arsenic insecticides against the cowpea curculio. J. Econ. Entomol. 31: 309–313.

Bell, J.V. and R.J. Hamalle. 1970. Three fungi tested for control of cowpea curculio, *Chalcodermus aeneus*. J. Invertebr. Pathol. 15: 447–450.

Bell, J.V. and R.J. Hamalle. 1971. A bacterium and dipterous parasite in wild populations of cowpea curculio larvae—Effects of treatment with spores of *Metarrhizium-anisopliae*. J. Invertebr. Pathol. 17: 256–259.

Dara, S.K., T.A. Goble and D.I. Shapiro-Ilan. 2018. Leveraging the ecology of invertebrate pathogens in microbial control, *In* Hajek, A.E. and D.I. Shapiro-Ilan (eds.). Ecology of Invertebrate Diseases. John Wiley & Sons.

Goettel, M.S. and D. Inglis. 1997. Fungi: Hyphomycetes, Pp. 213–249. *In* Manual of Techniques in Insect Pathology. Academic Press, San Diego, CA.

Hothorn, T., F. Bretz and P. Westfall. 2008. Simultaneous inference in general parametric models. Biometr. J. 50: 346–363.

Jaramillo, J., C. Borgemeister, L. Ebssa, A. Gaigl, R. Tobon and G. Zimmermann. 2005. Effect of combined applications of *Metarhizium anisopliae* (Metsch.) Sorokin (Deuteromycotina : Hyphomycetes) strain CIAT 224 and different dosages of imidacloprid on the subterranean burrower bug *Cyrtomenus bergi* Froeschner (Hemiptera : Cydnidae). Biol. Contr. 34: 12–20.

Kaya, H.K. and S.P. Stock. 1997. Techniques in Insect Nematology, Academic Press, San Diego, CA.

Lacey, L.A. and D.I. Shapiro-Ilan. 2008. Microbial control of insect pests in temperate orchard systems: Potential for incorporation into IPM. Annu. Rev. Entomol. 53: 121–144.

Lewis, E.E. and D.J. Clarke. 2012. Nematode parasites and entomopathogens, Pp. 395–424. *In* F.E. Vega and H.K. Kaya, eds. Insect Pathology, 2nd ed. Elsevier, Amsterdam, The Netherlands.

Mannion, C.M. and R.K. Jansson. 1992. Comparison of 10 entomopathogenic nematodes for control of sweet-potato weevil (Coleoptera, Apionidae). J. Econ. Entomol. 85: 1642–1650.

Nagesh, M., N.K.K. Kumar, A.N. Shylesha, N. Srinivasa, S. Javeed and R. Thippeswamy. 2016. Comparative virulence of strains of entomopathogenic nematodes for management

Downloaded From: https://bioone.org/journals/Journal-of-Entomological-Science on 19 Mar 2026
Terms of Use: https://bioone.org/terms-of-use Access provided by Clemson University

of eggplant grey weevil, *Myllocerus subfasciatus* Guerin (Coleoptera: Curculionidae). Indian J. Exp. Biol. 54: 835–842.

Nguessan, K.F. and R.B. Chalfant. 1990. Dose-response of the cowpea curculio (Coleoptera: Curculionidae) from different regions of Georgia to some currently used pyrethroid insecticides. J. Entomol. Sci. 25: 219–222.

Pinheiro, J.C. and D.M. Bates. 2000. Mixed-effects Models in S and S-PLUS, Springer, New York.

Poinar, G.O., Jr., R. Gaugler and H.K. Kaya. 1990. Biology and taxonomy of Steinernematidae and Heterorhabditidae. CRC Press, Boca Raton, FL.

Poinar, G.O., Jr. and P.S. Grewal. 2012. History of entomopathogenic nematology. J. Nematol. 44: 153–161.

Quintella, E.D. and D.W. Roberts. 1992. Control of *Chalcodermus bimaculatus* (Boheman) (Coleoptera: Curculionidiae) on soil with *Beauveria bassiana* (Bals.) Vuillemin and *Metarhizium anisopliae* (Metsch) Sorokin. Pesq. Agropec. Bras. 27: 95–105.

R Core Team 2016. R: A language and environment for statistical computing. R Core Team, Vienna, Austria. http://www.r-project.org/.

Riley, D.G., T. Coolong, R. Gitaitis, B. Dutta, A. Sparks, T. Grey, J. Schmidt, E. Fonsah and M. Toews. 2017. Crop profile for cowpeas in Georgia. Sponsored by Southern Integrated Pest Management Center, funded by USDA National Institute of Food and Agriculture. Coop Ext. Ser. Bull: B1480. http://extension.uga.edu/publications/detail.

Riley, D.G., A. Sparks and J. Kicklighter. 2014. Cowpea curculio in Southern Pea. University of Georgia Extension. http://extension.uga.edu/publications/detail.

Shapiro-Ilan, D.I. 2001. Virulence of entomopathogenic nematodes to pecan weevil (Coleoptera : Curculionidae) adults. J. Entomol. Sci. 36: 325–328.

Shapiro-Ilan, D.I. and T.E. Cottrell. 2006. Control of pests in pecan and peach using entomopathogenic nematodes: Challenges and successes. J. Nematol. 38: 292.

Shapiro-Ilan, D.I., T.E. Cottrell and B.W. Wood. 2011a. Effects of combining microbial and chemical insecticides on mortality of the pecan weevil (Coleoptera: Curculionidae). J. Econ. Entomol. 104: 14–20.

Shapiro-Ilan, D.I., W.A. Gardner, J.R. Fuxa, B.W. Wood, K.B. Nguyen, B.J. Adams, R.A. Humber and M.J. Hall. 2003. Survey of entomopathogenic nematodes and fungi endemic to pecan orchards of the southeastern United States and their virulence to the pecan weevil (Coleoptera : Curculionidae). Environ. Entomol. 32: 187–195.

Shapiro-Ilan, D.I., R.C. Han and C. Dolinksi. 2012. Entomopathogenic nematode production and application technology. J. Nematol. 44: 206–217.

Shapiro-Ilan, D.I., M. Jackson, C.C. Reilly and M.W. Hotchkiss. 2004. Effects of combining an entomopathogenic fungi or bacterium with entomopathogenic nematodes on mortality of *Curculio caryae* (Coleoptera : Curculionidae). Biol. Contr. 30: 119–126.

Shapiro-Ilan, D.I., T.C. Leskey and S.E. Wright. 2011b. Virulence of entomopathogenic nematodes to plum curculio, *Conotrachelus nenuphar*: Effects of strain, temperature, and soil type. J. Nematol. 43: 187–195.

Shapiro-Ilan, D.I. and C.W. McCoy. 2000. Susceptibility of *Diaprepes abbreviatus* (Coleoptera : Curculionidae) larvae to different rates of entomopathogenic nematodes in the greenhouse. Florida Entomol. 83: 1–9.

Slatten, B.H. and A.D. Larson. 1967. Mechanism of pathogenicity of *Serratia marcescens*. I. Virulence for the adult boll weevil. J. Invertebr. Pathol. 9: 78–81.

Smith, M.T., R. Georgis, A.P. Nyczepir and R.W. Miller. 1993. Biological-control of the pecan weevil, *Curculio caryae* (Coleoptera, Curculionidae), with entomopathogenic nematodes. J. Nematol. 25: 78–82.

Stark, J.E.P. and L.A. Lacey. 1999. Susceptibility of western cherry fruit fly (Diptera : Tephritidae) to five species of entomopathogenic nematodes in laboratory studies. J. Invertebr. Pathol. 74: 206–208.

Sudbrink, D.L., T.P. Mack and G.W. Zehnder. 1998. Alternate host plants of cowpea curculio (Coleoptera : Curculionidae) in Alabama. Florida Entomol. 81: 373–383.

Downloaded From: https://bioone.org/journals/Journal-of-Entomological-Science on 19 Mar 2026
Terms of Use: https://bioone.org/terms-of-use Access provided by Clemson University

Williams, C.D., A.B. Dillon, D. Ennis, R. Hennessy and C.T. Griffin. 2015. Differential susceptibility of pine weevil, *Hylobius abietis* (Coleoptera: Curculionidae), larvae and pupae to entomopathogenic nematodes and death of adults infected as pupae. Biocontrol 60: 537–546.

Wraight, S.P. and M.E. Ramos. 2002. Application parameters affecting field efficacy of *Beauveria bassiana* foliar treatments against Colorado potato beetle *Leptinotarsa decemlineata*. Biol. Control 23: 164–178.

Downloaded From: https://bioone.org/journals/Journal-of-Entomological-Science on 19 Mar 2026
Terms of Use: https://bioone.org/terms-of-use Access provided by Clemson University